



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2023 - 2024)

JEE(Advanced)
FULL SYLLABUS
04-05-2024

JEE(Main + Advanced) : ENTHUSIAST & LEADER COURSE (SCORE ADVANCED)

Time : 3 Hours

PAPER-2 (OPTIONAL)

Maximum Marks : 192

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **24** pages and that all the **17** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140

- | | |
|------------------------------------|--|
| • Boltzmann constant | $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$ |
| • Coulomb's law constant | $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ |
| • Universal gravitational constant | $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$ |
| • Speed of light in vacuum | $c = 3 \times 10^8 \text{ ms}^{-1}$ |
| • Stefan-Boltzmann constant | $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$ |
| • Wien's displacement law constant | $b = 2.89 \times 10^{-3} \text{ m-K}$ |
| • Permeability of vacuum | $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$ |
| • Permittivity of vacuum | $\epsilon_0 = \frac{1}{\mu_0 c^2}$ |
| • Planck constant | $h = 6.63 \times 10^{-34} \text{ J-s}$ |

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I (i) : (Maximum Marks: 28)

- This section contains **SEVEN (07)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

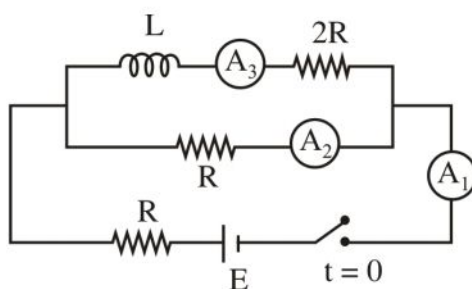
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. Figure shows a circuit consisting of an inductor and resistors connected with a cell. All ammeter are ideal. The switch was closed at $t = 0$. Choose the correct statement(s).



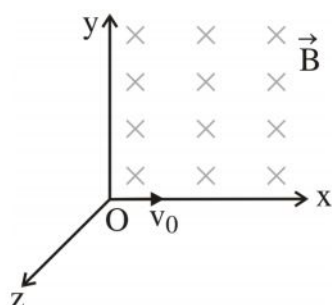
(A) Reading of A_1 is sum of reading of A_2 and A_3 .

(B) Reading of A_2 will be $\frac{\epsilon}{10R} \left(4 + e^{-\frac{5Rt}{2L}} \right)$.

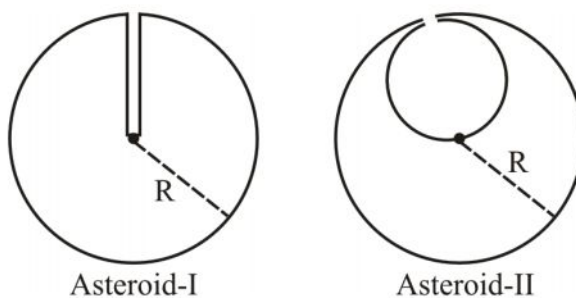
(C) Reading of A_3 will be $\frac{\epsilon}{5R} \left(1 - e^{-\frac{5Rt}{2L}} \right)$.

(D) In steady state reading of A_1 will be $\frac{\epsilon}{2R}$.

2. The ground state and first excited state energies of hydrogen atom are -13.6 eV and -3.4 eV respectively. If potential energy in ground state is taken to be zero, then :-
- (A) Potential energy in the first excited state would be 20.4 eV.
 (B) Total energy in the first excited state would be 23.8 eV.
 (C) Kinetic energy in the first excited state would be 3.4 eV.
 (D) Total energy in the ground state would be 13.6 eV.
3. A particle of mass m , charge q is projected from origin with velocity $v_0 \hat{i}$ in a region where uniform magnetic field $\vec{B} = B_0(-\hat{k})$ exist. If it experience drag force equal to $\vec{F} = -K\vec{v}$, where K is a positive constant and $\vec{v}$ is velocity vector, then :-

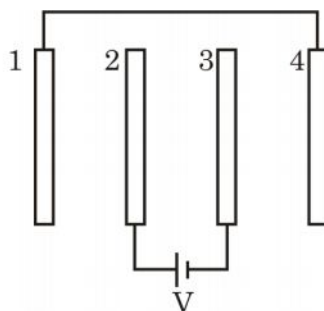


- (A) Distance travelled by particle before its velocity is along the positive x-axis is $\frac{mv_0}{K} \left(1 - e^{-\frac{2\pi K}{qB}} \right)$.
 (B) Distance travelled by particle before its velocity is along the positive x-axis is $\frac{2mv_0}{K} \left(1 - e^{-\frac{2\pi K}{qB}} \right)$.
 (C) Speed of the particle decreases exponentially with time.
 (D) Particle finally stops at origin.
4. Consider two solid spherical asteroid of uniform density of mass M and radius R . In one asteroid a tunnel of very small size of depth R is bored to the centre and in other asteroid a spherical cavity of radius $\frac{R}{2}$ is made as shown in the figure. Now, identical particles of mass m are dropped into cavities of both asteroids from top most point P . The force experienced by particles are F_I and F_{II} respectively in cavities of asteroid I and II, when they are x distance away from centre of asteroids. If the time taken by particles to reach the centre of asteroids is T_I and T_{II} respectively then



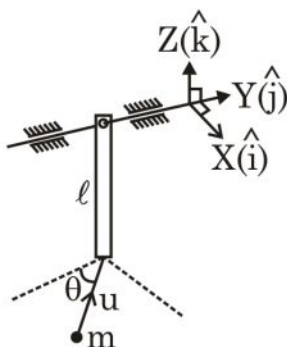
- (A) The ratio of F_I/F_{II} is equal to $2x/R$.
 (B) The ratio of F_I/F_{II} is equal to x/R .
 (C) The ratio of T_I/T_{II} is equal to $\pi/4$.
 (D) The ratio of T_I/T_{II} is equal to $\pi/2$.

5. Four identical conducting plates of area A each are connected as shown in circuit and the gap ' d ' between adjacent plates is same. Initially charge on each plate is zero. Select the correct statements.



- (A) Equivalent capacitance between the terminals of battery is $\left(\frac{3\epsilon_0 A}{2d}\right)$.
- (B) Sum of charges on plate-1 and plate-4 is zero.
- (C) The charge on right side of plate-2 is $\left(-\frac{\epsilon_0 A V}{d}\right)$.
- (D) The charge on left side of plate-2 is zero.
6. Train is moving with constant speed along a circular track. If length of the train is one fourth of length of circular track then which of the following is/are correct options? (Assume that sound source is at engine).
- (A) Frequency observed by a passenger who is sitting in the middle of train (equidistance from front and rear end) will continuously increase.
- (B) Frequency observed by a passenger who is sitting in the middle of train (equidistance from front and rear end) will remain constant but more than actual frequency.
- (C) Frequency observed by passenger who is sitting in the middle of train (equidistance from front and rear end) will remains constant and equal to actual frequency.
- (D) Wavelength observed by the person who is on the rear end of train is more than the actual wavelength of sound wave.

7. A particle of mass m moving in XY horizontal plane strikes the end of the vertical rod at an angle 37° with Y-axis. Rod has mass m and length ℓ and it is free to rotate about hinge along Y-axis. If particle strikes rod with speed u and sticks to it, find the **CORRECT** choice(s).



- (A) Velocity of particle just after impact will be $-\frac{4u\hat{i}}{5} + \frac{9u\hat{j}}{5}$.
- (B) Velocity of particle just after impact will be $-\frac{9u\hat{i}}{20} + \frac{4u\hat{j}}{10}$.
- (C) Minimum speed so that rod completes full rotation about hinge is $\frac{10}{3}\sqrt{2g\ell}$.
- (D) Angular momentum of system during collision is conserved about Y-axis.

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct option is chosen.

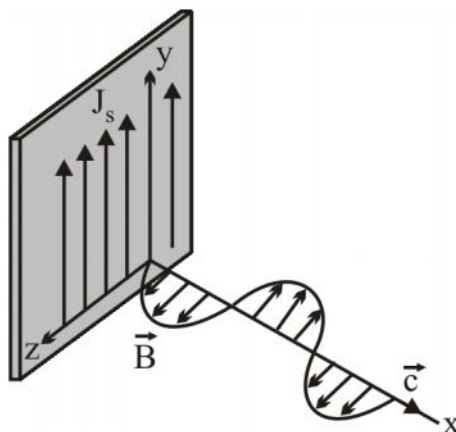
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -1 In all other cases.

Paragraph for Questions 8 and 9

A large, flat sheet carries a uniformly distributed electric current with current per unit width J . This current creates a magnetic field on both sides of the sheet, parallel to the sheet and perpendicular to the current, with magnitude $B = \frac{1}{2} \mu_0 J_s$. If the current is in the y direction and oscillates in time according to $J_{\max} (\cos \omega t) \hat{j}$, the sheet radiates an electromagnetic wave. Figure shows such a wave emitted from one point on the sheet chosen to be the origin. Such electromagnetic waves are emitted from all points on the sheet. The magnetic field of the wave to the right of the sheet is described by the wave function

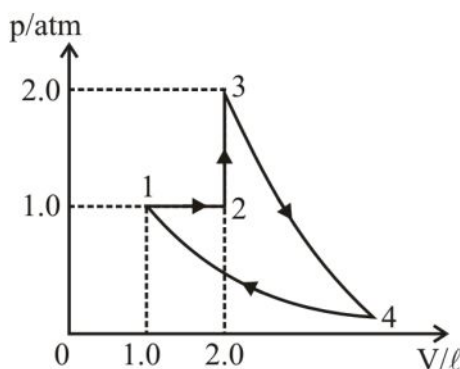
$$\vec{B} = \frac{1}{2} \mu_0 J_{\max} [\cos(kx - \omega t)] \hat{k}$$



8. The electric field of wave to the right of the sheet is :-
- (A) $\frac{1}{2} \mu_0 c J_{\max} \cos(kx - \omega t) \hat{j}$ (B) $-\frac{1}{2} \mu_0 c J_{\max} \cos(kx - \omega t) \hat{j}$
- (C) $\frac{1}{2} \frac{\mu_0 J_{\max}}{c} \cos(kx - \omega t) \hat{k}$ (D) $-\frac{1}{2} \frac{\mu_0 J_{\max}}{c} \cos(kx - \omega t) \hat{k}$
9. If the intensity of radiation emitted by the sheet in each direction is 570 W/m^2 , the maximum value of sinusoidal current density is :-
- (A) 1.28 A/m (B) 2.46 A/m
- (C) 3.48 A/m (D) 4.92 A/m

Paragraph for Questions 10 and 11

3.2g of oxygen starts from the initial state 1 ($p_1 = 1.0 \text{ atm}$, $V_1 = 1.0 \text{ L}$), reaches state 2 ($V_2 = 2.0 \text{ L}$) through an isobaric process, and then passes through the isochoric process. The process reaches state 3 ($p_3 = 2.0 \text{ atm}$), and then reaches state 4 through an adiabatic process. The temperature of state 4 is exactly the same as the temperature of state 1. Finally, it returns to state 1 through an isothermal process, forming a cyclic process.



10. The efficiency of cyclic process is :-

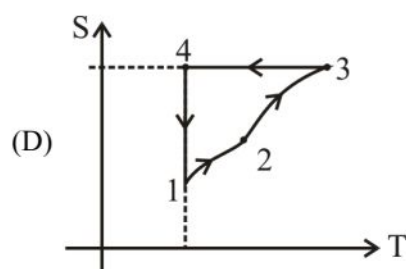
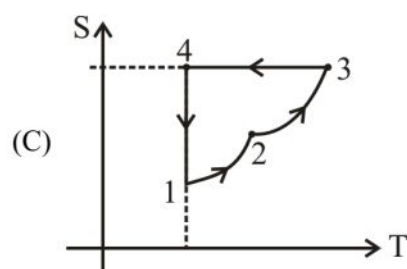
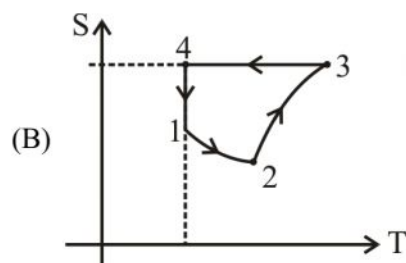
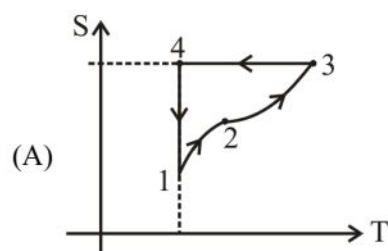
(A) $1 - \frac{\ell n 2}{2}$

(B) $1 - \frac{3 \ell n 2}{9}$

(C) $1 - \frac{6 \ell n 2}{13}$

(D) $1 - \frac{12 \ell n 2}{17}$

11. The variation of entropy with temperature is given by :-



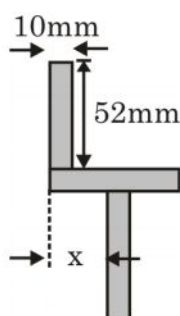
SECTION-II : (Maximum Marks: 24)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

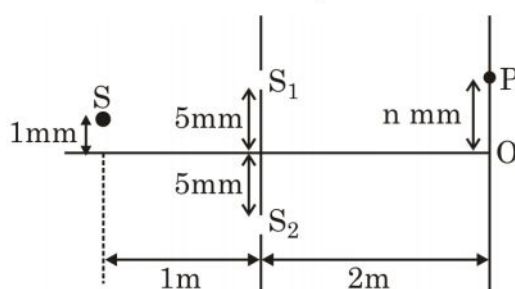
Full Marks : +4 If **ONLY** the correct numerical value is entered.

Zero Marks : 0 In all other cases.

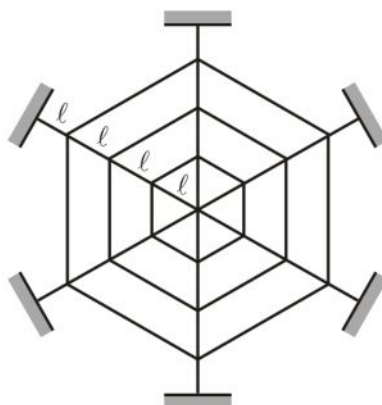
- Three identical uniform dominos are arranged as shown in the diagram. The possible values of x such that the dominoes are in stable equilibrium is $\alpha < x < \beta$. α , β and x are measured in mm. Then the value of $\alpha + \beta$ (in mm) is :-



- A magnetic disc is moving on an air-cushioned table above a metal sheet. Due to the generated eddy currents there is a retarding force exerted on the disc, which is proportional to the speed of the disc. Moving above a sheet made of aluminum the disc stops after covering a distance of 30 cm, but when the sheet is made of copper the disc stops after covering only 20 cm. How much total distance (in cm) will the disc cover if first it travels above a piece of copper sheet of width 15 cm, and then continues its motion above an aluminum sheet? (The initial speeds of the disc are the same in all three cases.)
- In the young's double slit experiment a point source of $\lambda = 5000 \text{ \AA}$ is placed slightly off the central axis as shown in the figure. If 70th order maximum is formed at P, the value of n is :-



4. A rocket is launched normal to the surface of the Earth, away from the Sun along the line joining the Sun and the Earth. The Sun is 3×10^5 times heavier than the Earth and is at a distance 2.5×10^4 times larger than the radius of the Earth. The escape velocity from Earth's gravitational field is $v_e = 11.2 \text{ km s}^{-1}$. The minimum initial velocity (v_s) (in km/sec) required for the rocket to be able to leave the Sun-Earth system is closest to :- (Ignore the rotation and revolution of the Earth and the presence of any other planet)
5. Troposphere is the lowest layer of atmosphere, whose temperature decreases linearly with height (y) according to $T = T_0 - \alpha y$, where T_0 is temperature at Earth's surface and the constant α is called lapse rate of temperature. The average molar mass of normal atmospheric air is M_0 kg/mole and for the height of troposphere, the gravity field g is constant. The atmospheric pressure at ground is P_0 and gas constant is R . The atmospheric pressure as a function of height y will be $P(y) = P_0 \left(1 - \frac{\alpha y}{T_0} \right)^{\frac{n M_0 g}{2 \alpha R}}$, then value of n is :-
6. The web of negligible weight has the shape shown in the picture. The hexagons are regular and divide the radial fibers into equal parts of length ℓ . The ends of the web are fixed at the same height above the ground. At the beginning, the web is not sagged. When a spider of mass m moves to the center of the web, the web bends so that its center moves closer to the ground by a distance h . Determine the Young's modulus of elasticity Y of a spider fiber that has a diameter d . If the answer is $x \times 10^{17} \text{ Pa}$, fill the value of x . (Given : $\ell = 0.10 \text{ m}$, $m = 0.010 \text{ kg}$, $h = 5.0 \text{ mm}$, $d = 0.15 \text{ } \mu\text{m}$)



PART-2 : CHEMISTRY
SECTION-I (i) : (Maximum Marks: 28)

- This section contains **SEVEN (07)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
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Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

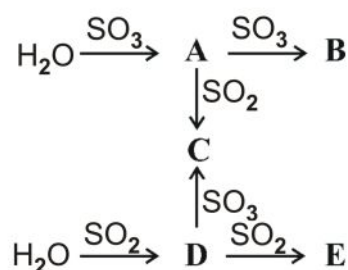
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

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- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. The chain of reaction is completely hypothetical and based upon formula matching only.
Consider 1 molecule of all the compounds H_2O , SO_3 , SO_2 , A, B, C, D, E in the given chain reaction



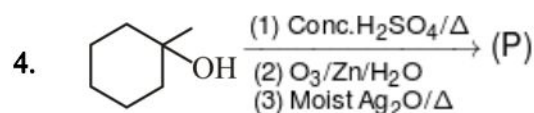
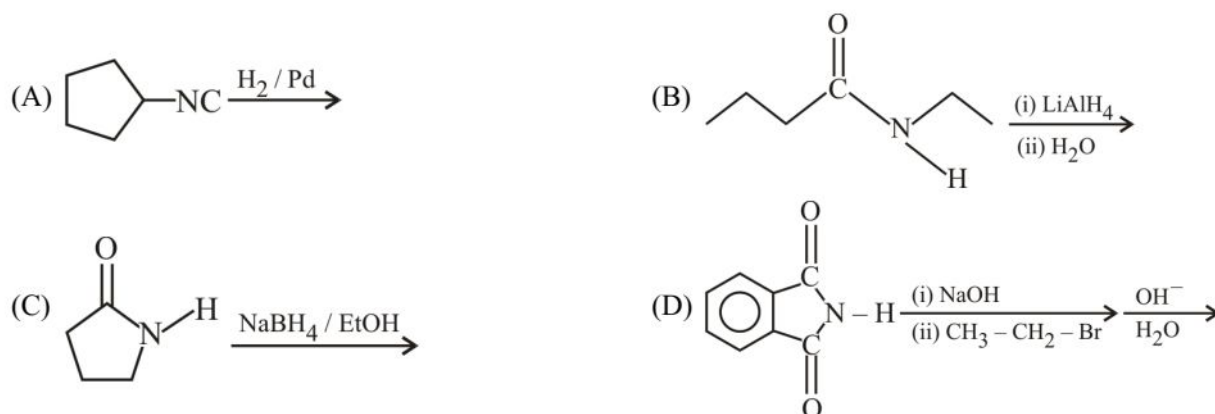
Select the **CORRECT** statement :

- (A) When one oxygen atom is removed from formula of B then formula of C is formed
- (B) C has S – S linkage in it's structure
- (C) E has S – O – S linkage in it's structure.
- (D) C is $\text{H}_2\text{S}_2\text{O}_6$ and E is $\text{H}_2\text{S}_2\text{O}_5$

2. Co-ordination entity Wave length of light absorbed (nm)
- | | |
|---|---|
| $[\text{CoCl}(\text{NH}_3)_5]^{2+}$ | P |
| $[\text{Co}(\text{NH}_3)_5(\text{H}_2\text{O})]^{3+}$ | Q |
| $[\text{Co}(\text{NH}_3)_6]^{3+}$ | R |
| $[\text{Co}(\text{CN})_6]^{3-}$ | S |
| $[\text{Cu}(\text{H}_2\text{O})_4]^{2+}$ | T |

Select the correct orders.

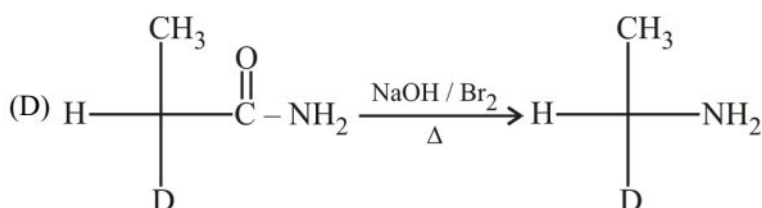
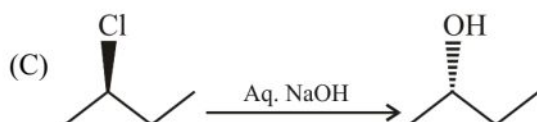
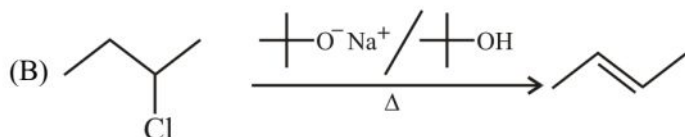
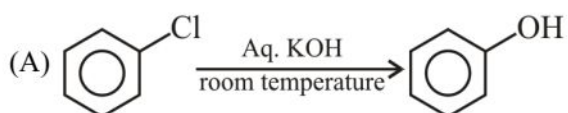
- (A) $P > R > Q$
 (B) $T > P > Q > R > S$
 (C) $P > Q > R$
 (D) $T > Q > P > R > S$
3. Which reactions/s can produce 2°-amines as major product?



Correct statement(s) for above reaction sequence :

- (A) Product (P) shows Tollen's test.
 (B) Product (P) shows Iodoform test.
 (C) Above reaction sequence involve elimination reaction.
 (D) Product (P) can show Lucas test

5. Which of the following reaction/s is/are correctly represented with major products?



6. $P_A = (235y - 125xy)$ mm of Hg. P_A is partial pressure of A, x is mole fraction of B in liquid phase in the mixture of two liquids A and B and y is the mole fraction of A in vapour phase, then :

(A) $P_A^\circ = 110$

(B) $P_B^\circ = 110$

(C) $P_A^\circ = 235$

(D) $P_B^\circ = 235$

7. The correct information(s) is/are :

(A) The minimum concentration of an electrolyte in millimoles per litre required to cause precipitation of a sol in two hours is called coagulation power

(B) Smaller the coagulation value, higher will be the coagulation power of that ion.

(C) Lyophilic sols may be coagulated by adding an electrolyte or a suitable solvent.

(D) Lyophobic colloids used to protect lyophilic colloids are called protective colloids.

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct
- For each question, choose the option corresponding to the correct answer.
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Full Marks : +3 If **ONLY** the correct option is chosen.

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Negative Marks : -1 In all other cases.

Paragraph for Questions 8 and 9

The electrochemical cell shown below is a concentration cell.

$M | M^{2+}$ (saturated solution of a sparingly soluble salt, MX_2) $|| M^{2+}$ ($0.001 \text{ mol dm}^{-3}$) $| M$

The emf of the cell depends on the difference in concentrations of M^{2+} ions at the two electrodes. The emf of the cell at 298 K is 0.059V.

8. The value of ΔG (kJ mol^{-1}) for the given cell $M(s) + M^{2+}(aq) \rightarrow M(s) + M^{2+}(aq)$ is (take $F = 96500 \text{ C mol}^{-1}$)
- (A) -5.7
(B) 5.7
(C) 11.4
(D) -11.4
9. The solubility product (K_{sp} ; $\text{mol}^3 \text{ dm}^{-9}$) of MX_2 at 298 K based on the information available for the given concentration cell is $x \times 10^{-15}$ (take $2.303 \times R \times 298/F = 0.059 \text{ V}$). The value of x is ?
- (A) 1×10^{-15}
(B) 4×10^{-15}
(C) 1×10^{-12}
(D) 4×10^{-12}

Paragraph for Questions 10 and 11

If values of quantum numbers can be considered as follows.

$$n = 1, 2, 3 \dots \infty$$

$$\ell = 0 \text{ to } (n - 1)$$

$$m = -\ell \text{ to } +\ell \text{ (integral value including zero)}$$

$$s = -\frac{1}{4}, -\frac{1}{2}, +\frac{1}{2}, +\frac{1}{4}$$

10. According to above consideration, the maximum number of e^- associated to 2nd shell is :
- (A) 64
(B) 32
(C) 16
(D) 8
11. According to above consideration, the maximum number of e^- associated to a p-orbital is :
- (A) 2
(B) 4
(C) 6
(D) 12

SECTION-II : (Maximum Marks: 24)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)

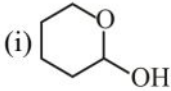
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

1. How many of the following cations gives ppt with H_2S in acidic solution :-
 Zn^{2+} , Cu^{2+} , Ni^{2+} , Sr^{2+} , Al^{3+} , Bi^{3+} , Sn^{2+} , Pb^{2+}
2. Among the following statements
 - (i) Ag_2SO_3 is insoluble in dilute HNO_3 .
 - (ii) Nitrogen dioxide (NO_2) is absorbed by ferrous sulphate solution forming brown solution.
 - (iii) Heavy metal chlorides like $AgCl$, $HgCl_2$. etc. do not respond to chromyl chloride test.
 - (iv) The group reagent for III group basic radicals is $NaOH$.
 - (v) Green colour of alkaline tetrahydroxidochromate (III) becomes yellow on adding H_2O_2 .
 - (vi) Titan yellow solution gives blue colouration/ppt. with a acidic solution containing Mg^{2+} ions
 - (vii) Solution of nitrite is decomposed by sulphamic acid.
 - (viii) Fe^{2+} ions give brown colour precipitate with $[Fe(CN)_6]^{3-}$ ions solution.
 - (ix) Green precipitate of $Cr(OH)_3$ is soluble in excess Na_2O_2 .

If "**a**" is total number of correct while "**b**" is total number of incorrect statements find $\frac{a^2 + b^2}{3}$.

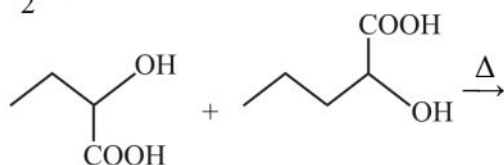
3. (i)  (ii) $\text{CH}_3 - \overset{\text{O}}{\underset{\parallel}{\text{C}}} - \text{H}$ (iii) $\text{CH}_3 - \overset{\text{O}}{\underset{\parallel}{\text{C}}} - \text{CH}_3$
- (iv) $\text{CH}_3 - \overset{\text{O}}{\underset{\parallel}{\text{C}}} - \underset{\text{OH}}{\underset{|}{\text{CH}}} - \text{CH}_3$ (v) $\text{CH}_3 - \text{C} \equiv \text{C} - \text{CH}_3$
- (vi) $\text{H} - \overset{\text{O}}{\underset{\parallel}{\text{C}}} - \text{OH}$ (vii)  (viii) $\text{CH}_3 - \text{CH}_2 - \text{OH}$ (ix) $\text{CH}_3 - \overset{\text{O}}{\underset{\parallel}{\text{C}}} - \text{H}$

Number of compounds that gives Tollen's test = x.

Number of compounds that gives Fehling's test = y.

Then $x + y = \underline{\hspace{2cm}}$

4. 'X' is the number of possible products (including stereoisomers) in the following reaction, then value of $\frac{X}{2}$ is



5. Fixed amount of an ideal gas contained in a sealed rigid vessel ($V = 24.6$ litre) at 1.0 bar is heated reversibly from 27°C to 127°C . Determine change in Gibb's energy (in J), if entropy of gas $S = 10 + 10^{-2} T$ (J/K)
6. The solubility of a salt (AB) of weak acid at $\text{pH} = 3$ is $\sqrt{Y} \times 10^{-3} \text{ mol L}^{-1}$. The value of Y is _____.
(Given that the value of solubility product of AB (K_{sp}) = 2×10^{-10} and the value of ionization constant of $\text{HB}(K_{\text{a}}) = 1 \times 10^{-8}$)

PART-3 : MATHEMATICS
SECTION-I (i) : (Maximum Marks: 28)

- This section contains **SEVEN (07)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but ONLY three options are chosen.

Partial Marks : +2 If three or more options are correct but ONLY two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. Let A, B, C be 3 sets of complex numbers as defined below :

$$A = \{z : |z + 1| \leq 2 + \operatorname{Re}(z)\}, B = \{z : |z - 1| \geq 1\} \text{ and } C = \left\{z : \left|\frac{z-1}{z+1}\right| \geq 1\right\}$$

Then which of the following are correct :

- (A) Area of the region bounded by $A \cap B \cap C$ is $\sqrt{3}$
- (B) Area of the region bounded by $A \cap B \cap C$ is $2\sqrt{3}$
- (C) The real part of the complex number in region $A \cap B \cap C$ having maximum amplitude and modulus is $-\frac{3}{2}$
- (D) The number of point having integral coordinates in region $A \cap B \cap C$ is 7
2. If A is the area enclosed between the curves $y = \ln(x + e)$, $x = \ln\left(\frac{1}{y}\right)$ and x-axis then :
- (A) A is prime number
- (B) A is composite number
- (C) A is irrational number
- (D) A^2 is even integer

3. The solution of the differential equation $3 \frac{dx}{dy} = \frac{x}{x^3 - y}$ is $x^\ell = mx^n y + c$, then which of the following options are correct :
- (A) $\ell + m + n = 11$
- (B) $\ell + n = 9$
- (C) $\ell + 2m = 10$
- (D) $m + n = 4$
4. The value of $\sum_{n=2}^{\infty} \frac{n^4 + 3n^2 + 10n + 10}{2^n(n^4 + 4)}$ is less than :
- (A) $\frac{11}{7}$
- (B) $\frac{11}{8}$
- (C) $\frac{11}{9}$
- (D) $\frac{11}{10}$
5. Tangent drawn at point P(2, -5) of a parabola intersects its tangent at vertex at (1, -3). If (8, 3) is a point on SP, where S is focus of the parabola then
- (A) focus of the parabola is (5, -1)
- (B) Directrix of the parabola is $x - y - 2 = 0$
- (C) Slope of normal to parabola at other end of focal chord drawn from point P is -2
- (D) Circle drawn on PS as diameter touches the straight line $x - y - 4 = 0$
6. Consider the equation $(x^2 - 4x + 6)^2 - (k + 2)(x^2 - 4x + 6) + k + 5 = 0$ ($k \in \mathbb{R}$). Which of the following are correct :
- (A) there is no integral value of k for which given equation has exactly 4 real solutions.
- (B) there is only one integral value of k for which given equation has exactly 3 real solutions.
- (C) there is only one integral value of k for which given equation has exactly one real solution.
- (D) there are only 3 positive integral values of k for which given equation has no real solution.
7. Consider the limit $\lim_{x \rightarrow 0} \frac{a \tan x + b \tan^{-1} x + c \sin^{-1} x}{x^5} = 3$, where a, b, c $\in \mathbb{R}$, then correct option/s is/are
- (A) $a = 30$
- (B) $b = 10$
- (C) $c = -40$
- (D) $b = -5$

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -1 In all other cases.

Paragraph for Questions 8 and 9

$$\text{Let } f(x) = \begin{cases} x^2 - 3x + 2 & 0 \leq x \leq 3 \\ |x - 5| & 3 < x \leq 8 \\ \log_2(12 - x) & 8 < x \leq 11 \end{cases}$$

8. If $f(x) = k$ has 5 distinct solutions then sum of integral values of k is
- (A) 1
(B) 3
(C) 5
(D) 7
9. Let ℓ = maximum number of distinct roots of $|f(x)| = k$ and $m = \lim_{x \rightarrow 3} [f(3)^{f(x)}]$, then $\ell + m$ is (Where $[.]$ denotes greatest integer function)
- (A) 11
(B) 9
(C) 7
(D) 5

Paragraph for Questions 10 and 11

Let $\frac{d}{dx}(f(x)) = \frac{e^{\sin x}}{x}$, $x > 0$, then answer the following :

10. If $\int_1^4 \frac{2e^{\sin x^2}}{x} dx = f(k) - f(1)$ and $xf''(x) + f'(x) = g(x)$, then $k + \left[g\left(\frac{\pi}{2}\right)\right]$ is
- (A) 9
- (B) 16
- (C) 27
- (D) 36
11. Let $I = \int_0^{\pi/2} x \cos x f'(x) dx = e^\lambda - 1$ and $xf'(x) = T(x)$, then $[\lambda] + T\left(\frac{\pi}{2}\right)$ is equal to : (Where $[.]$ represents G.I.F.)
- (A) $1 + e$
- (B) 3
- (C) 2
- (D) 4

SECTION-II : (Maximum Marks: 24)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If ONLY the correct numerical value is entered.
Zero Marks : 0 In all other cases.

1. If k is the number of four digit natural numbers, such that repetition of digits is not allowed and absolute difference between its first and last digit is 2 then $\frac{k}{480}$ is :
2. If a is the minimum value of x_1 and b is the maximum value of x_2 such that $\int_{x_1}^{x_2} (x^4 - 2x^2) dx$ is minimum then the area of region enclosed between the curves $y = x^4 - 2x^2$, the x -axis and the ordinates $x = a$, $x = b$ is :
3. If a variable circle touches the straight line $y = x$ and cuts the chord of length $2\sqrt{2}$ from $x + y = 0$ then the length of latus rectum of the locus of the centers of the circles is :
4. If the value of $\lim_{n \rightarrow \infty} n \left(\frac{1}{3n^2 + 8n + 4} + \frac{1}{3n^2 + 16n + 16} + \frac{1}{3n^2 + 24n + 36} + \dots \right)$ is $\frac{1}{4} \ln \left(\frac{n}{k} \right)$ where $\gcd(n, k) = 1$ then $\frac{n}{k}$ is :
5. For two data sets each of the size 5, variance are given to be 4 and 5 respectively and mean are 2, 4 respectively, the variance of combined data is equal to
6. Let A is a 2×2 matrix such that $AA^T = I$, where I is identity matrix then the maximum value of the sum of the absolute values of entries of matrix A is K then find $[K]$ where $[.]$ represents greatest integer function :

Space for Rough Work

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same.

Signature of the Invigilator

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E-24/24

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